Level 2 – Medium exercises

Focus: combined conditions, nested logic, loop control, validation and business rules.

No.	Exercise	Requirements
M01	Progressive Electricity Bill	Apply three unit-based tariff slabs and calculate the cumulative bill.
M02	Employee Overtime Calculator	Calculate basic and overtime pay using a 40-hour workweek and 1.5x overtime rate.
M03	Leap Year Detector	Identify leap years using Gregorian calendar rules.
M04	Assessment Grade Calculator	Accept a score, validate it and assign the correct grade.
M05	Secure Login Attempt Simulator	Allow three attempts, stop immediately on success and report exhausted attempts.
M06	ATM Withdrawal Simulator	Check balance, withdrawal validity, denomination and available funds.
M07	Divisibility Analyzer	Display numbers divisible by 3, 5 or both within a user-supplied range.
M08	Prime Number Detector	Determine whether a positive integer greater than one is prime.
M09	Assessment Analytics	Calculate average score, number of passes, fails and pass percentage.
M10	Menu-Driven Calculator	Support addition, subtraction, multiplication and division until the user exits.

Level 3 – Hard / Tricky exercises

Advanced problem solving

These five questions should be treated as analytical coding challenges rather than straightforward syntax exercises.

No.	Exercise	Requirements
H01	Second Largest Distinct Number	Find the second-largest distinct value among three inputs, handling duplicates.
H02	Number Guessing Challenge	Generate a random number and give the user a limited number of attempts, with higher/lower hints.
H03	Reverse an Integer	Reverse the digits mathematically without converting the number to a string.
H04	Sum of Digits	Calculate the digit sum of a positive or negative integer without string conversion.
H05	Prime Number Series	Generate every prime number up to N using nested loops and an optimized divisor boundary.

Level 4 – Integrated scenario challenge

H06 – Business scenario

Employee Training Certification and Performance Reporting

Business context

A company is conducting a technical training programme for 50 employees. HR wants to identify certification eligibility and generate a performance report.

Functional requirements

1. Accept the number of employees (maximum 500).
2. Capture employee name, attendance and assessment score.
3. Validate names and numeric ranges.
4. Require at least 75% attendance and 60 assessment marks for certification.
5. Classify performance into four grades.
6. Calculate average assessment score.
7. Identify the highest-performing employee.
8. Identify the lowest-performing employee.
9. Calculate eligibility percentage.
10. Count employees in each performance category.

Technical constraints

- Use variables, operators, conditions, loops and exception handling.
- Do not use lists, dictionaries, functions, classes or external libraries.
- Reject invalid attendance and marks rather than silently skipping them.
- Test boundary values and invalid inputs.
- Run Ruff and Bandit before submission.

Expected business output

```
EMPLOYEE TRAINING REPORT
------------------------
Total participants       : 50
Eligible participants    : 38
Not eligible             : 12
Excellent                : 10
Good                     : 15
Average                  : 18
Needs Improvement        : 7
Average assessment      : 74.65
Certification rate      : 76.00%
Top performer            : Employee 024
```